

Statistical Walkthrough

Figures and Code Explained v2 - 6 Comparisons with Corrected Data

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Purpose

This document explains the statistical concepts behind each figure and code block in the tissue-specific DEG analysis. It uses basic and intermediate statistical terminology, with callout boxes for key concepts. All 6 radiation comparisons are covered, plus the new cross-comparison analyses.

Part I: Figures Explained

Figure 1: Stacked Bar Chart - DEG Counts by Organ

What it shows: For each comparison, a stacked bar chart displays the number of tissue-specific DEGs in each of the 5 broad organs, colored by direction (blue = upregulated, orange = downregulated).

Statistical concept - Frequency distribution: A bar chart is a visual representation of a frequency distribution - it shows how many items (DEGs) fall into each category (organ). The height of each bar represents the count. The stacked colors show a secondary categorical variable (direction).

Basic Term: Frequency

Frequency is simply how many times something occurs. If Root has 99 DEGs, the frequency for Root is 99. The bar chart makes it easy to compare frequencies across categories at a glance.

What to look for: In radiation_effect, Seedling has the tallest bar (148 DEGs, almost all upregulated). In genotype_interaction, Flower and Seed_Silique dominate, and the bars are mostly orange (downregulated). This visual difference between comparisons is quantified by the chi-square tests in Part II.

Figure 2: Volcano Plots by Organ

What it shows: A 5-panel volcano plot (one per organ) for each comparison. The x-axis is log2 fold change (effect size), the y-axis is $-\log_{10}(\text{padj})$ (statistical significance). Each point is a gene. Colored points are DEGs (passing both thresholds); grey points are not significant.

Statistical concept - Effect size vs significance: A volcano plot visualizes the two dimensions of differential expression simultaneously. The x-axis shows the magnitude of change (effect size): genes far from center have large fold changes. The y-axis shows statistical significance: genes high on the plot have small p-values. The dashed lines mark the thresholds ($|\log_2\text{FC}| = 1$ and $\text{padj} = 0.05$).

Intermediate Term: Why both axes matter

A gene can be statistically significant (high on y-axis) but have a tiny effect size (near center on x-axis). This happens with large sample sizes where even small differences become 'significant.' The dual threshold (both axes) ensures we only call genes that are both significant AND biologically meaningful (2-fold change).

What to look for: In radiation_effect, Seedling and Leaf_Shoot show many points in the upper-right (upregulated DEGs with large effect sizes). In genotype_interaction, points cluster in the upper-left (downregulated). GCR comparisons have almost no colored points - very few genes pass both thresholds at these low doses.

Figure 3: Heatmap - Top 50 DEGs x 33 Sub-tissues

What it shows: For each comparison, the top 50 DEGs (by padj) are displayed as rows, with the 33 AtGenExpress sub-tissues as columns. Colors represent z-scored expression (blue = below average, red = above average). The left annotation shows direction (blue = up, orange = down).

Statistical concept - Z-score normalization: A z-score transforms each value to show how many standard deviations it is from the mean. $z = (x - \text{mean}) / \text{sd}$. This allows comparing expression patterns across genes with very different absolute expression levels. A z-score of +2 means the gene is expressed 2 standard deviations above its own average.

Basic Term: Standard Deviation

Standard deviation (SD) measures how spread out values are. If most values are close to the mean, SD is small. If values vary widely, SD is large. Z-scores use SD as a ruler: $z=1$ means '1 SD above average,' $z=-2$ means '2 SD below average.' This puts all genes on the same scale.

What to look for: Tissue-specific DEGs should show a block of red (high expression) in their assigned tissue and blue elsewhere. For radiation_effect, many top DEGs are seedling-specific (red in seedling columns). For genotype_interaction, the pattern is more mixed, reflecting the interaction effect rather than a simple tissue-specific response.

Figure 4: UpSet Diagram - Organ Overlaps

What it shows: An UpSet diagram visualizes set intersections. Each organ is a set of DEGs. The bottom-left bars show set sizes (total DEGs per organ). The main bar chart shows the size of each intersection - how many DEGs are shared between specific combinations of organs.

Statistical concept - Set intersection analysis: Unlike a Venn diagram (which becomes unreadable with more than 3 sets), an UpSet diagram scales to any number of sets. The connected dots below each bar indicate which sets are included in that intersection. A bar with a single filled dot represents genes unique to that organ; bars with multiple connected dots represent genes shared between those organs.

Intermediate Term: Set intersection

An intersection is the set of items that belong to all specified sets simultaneously. If 10 DEGs are in both Root and Seedling, the Root-and-Seedling intersection has 10 elements. Since each DEG is assigned to only one predominant organ, most intersections will be empty - the bars with single dots (unique to each organ) will dominate.

What to look for: Because each tissue-specific gene is assigned to only one predominant organ, the largest bars should be the single-organ sets (no overlaps). This confirms that the Tau-based tissue assignment is working correctly - genes are being assigned to distinct tissues.

Figure 5: Forest Plot - Tissue Enrichment Test

What it shows: For each comparison, a forest plot displays the DEG rate (%) for each organ (dots), with the overall mean as a dashed reference line. Significance markers (*, **, ***) indicate Bonferroni-corrected p-values from Fisher's exact tests. Colors indicate enrichment (green), depletion (orange), or not significant (grey).

Statistical concept - Proportions, confidence intervals, and odds ratios: The DEG rate is a proportion: (DEGs in organ) / (total genes in organ). Fisher's exact test compares this proportion to the rest of the genome. The odds ratio (OR) quantifies the difference: $OR > 1$ means the organ is enriched for DEGs, $OR < 1$ means depleted. The 95% confidence interval shows the range of plausible OR values.

Basic Term: Odds Ratio

An odds ratio compares the odds of an event in two groups. If Seedling has 148 DEGs out of 1,167 genes (12.7%) and the rest has 374 out of 9,616 (3.9%), the odds ratio is about 3.7. This means a tissue-specific Seedling gene is 3.7x more likely to be a DEG than a gene from any other organ.

What to look for: In radiation_effect, Seedling (12.7%, ***) and Leaf_Shoot (8.1%, *) are enriched; Root (3.1%, ***), Flower (3.4%, *), and Seed_Silique (3.4%, *) are depleted. In Cs137_100cGy, Root (2.4%, **) and Seedling (3.0%, **) are enriched. The pattern differs by radiation type, which the CMH test in Part II quantifies.

Cross-Comparison Figures

Figure cross1: Tissue Response Heatmap

What it shows: A heatmap of DEG rates (%) for each organ (rows) across all 6 comparisons (columns). Darker orange = higher DEG rate. The top annotation bar shows radiation type. Numbers in each cell are the exact DEG rate.

Statistical concept - Contingency table visualization: This heatmap is a visual representation of a 5x6 contingency table. Each cell is a proportion (DEG rate) that can be compared across rows (organs) and columns (comparisons). The color scale makes patterns visible that would be hard to see in a raw table of numbers.

What to look for

Seedling (row 2) has the darkest cell under radiation_effect (11.96%). GCR40 and GCR80 columns are nearly white (0% DEG rate). The Cs-137 columns show intermediate values with a dose gradient (10cGy < 100cGy). This visual pattern is what the CMH and chi-square tests quantify.

Figure cross2: Direction Comparison

What it shows: A faceted grouped bar chart with up/down DEG counts for each organ, across all 6 comparisons. Blue bars = upregulated, orange bars = downregulated.

Statistical concept - Conditional distributions: Each facet (panel) shows the conditional distribution of direction given organ for one comparison. Comparing these conditional distributions across facets reveals whether the direction pattern changes by radiation condition. The chi-square test for direction (Test 4) formally tests whether these conditional distributions differ.

Figure cross3: Dose-Response

What it shows: Line plots of DEG count vs dose for each organ, separated by radiation source (Cs-137 vs GCR). Each line is an organ, colored consistently with the organ color palette.

Statistical concept - Dose-response relationship: A dose-response curve shows how a biological outcome (DEG count) changes with increasing dose of a treatment (radiation). A steeper slope means greater sensitivity to dose. The CMH test for dose-response (OR=2.846 for Cs-137) formally tests whether the odds of being a DEG increase with dose, after accounting for tissue differences.

Figure cross4: Radiation Type Comparison

What it shows: A grouped bar chart comparing DEG rates by organ between Co-60 100 Gy and Cs-137 100 cGy. Blue bars = Co-60, green bars = Cs-137.

Statistical concept - Stratified comparison: This figure visualizes the comparison that the CMH test quantifies. The CMH test stratifies by organ (treating each organ as a separate stratum) and asks: after

accounting for organ-specific baseline rates, is there an overall difference in DEG rate between the two radiation types? The answer is yes ($OR=0.328$, $p=1.1e-39$): Co-60 produces 3.1x more tissue-specific DEGs than Cs-137.

Intermediate Term: Stratification

Stratification means analyzing each subgroup (organ) separately, then combining the results. This is more powerful than ignoring organ differences because it removes the confounding effect of organs having different baseline DEG rates. The CMH test is the stratified version of the chi-square test.

Part II: Code Blocks Explained

Code Block 1: Tau Tissue-Specificity Index

```
compute_tau <- function(expr_matrix) {
  tau <- apply(expr_matrix, 1, function(x) {
    x <- as.numeric(x)
    if (max(x) == 0) return(NA)
    sum(1 - x / max(x)) / (length(x) - 1)
  })
  return(tau)
}
```

What it does: Computes the Tau index for each gene across all tissues. The `apply()` function loops over rows (genes). For each gene, it normalizes expression by the maximum, subtracts from 1, sums, and divides by (n-1).

Statistical concept - Tissue-specificity metric: Tau is a tissue-specificity score ranging from 0 to 1. The formula $\tau = \sum(1 - x_i/\max(x)) / (n-1)$ measures how unevenly a gene is expressed across tissues. If a gene is expressed only in one tissue, all other values are near 0, so each $(1 - 0/\max) = 1$, and tau approaches 1. If expression is uniform, each $(1 - x/\max)$ is near 0, and tau approaches 0.

Basic Term: Normalization

Normalization means putting different things on the same scale. Here, dividing by $\max(x)$ normalizes all expression values to [0, 1], so genes with very different absolute expression levels can be compared. A gene expressed at 1000 in one tissue and 10 elsewhere gets the same Tau as one expressed at 100 and 1.

Code Block 2: Chi-Square Test of Independence

```
contig <- table(ts_genes$predominant_broad_organ, ts_genes$deg_flag)
chi_test <- chisq.test(contig)
```

What it does: Builds a contingency table (organ x DEG status) and runs a chi-square test of independence.

Statistical concept - Chi-square test of independence: This test asks: is the proportion of DEGs the same across all organs, or does it differ? The null hypothesis is that organ and DEG status are independent (no tissue enrichment). The test statistic X^2 measures how much the observed counts deviate from what we'd expect under independence. A large X^2 with a small p-value means the organs have significantly different DEG rates.

Intermediate Term: Expected vs Observed

The chi-square test compares observed counts to expected counts. If organs and DEG status were independent, we'd expect each organ to have the same DEG rate (the overall average). The test measures how far the actual data deviates from this expectation. $X^2 = \sum((\text{observed} - \text{expected})^2 / \text{expected})$.

Results: radiation_effect: $X^2=233.6$, $p=2.2e-49$ (highly significant). genotype_interaction: $X^2=14.4$, $p=0.006$ (significant). Cs137_100cGy: $X^2=53.9$, $p=5.7e-11$. Cs137_10cGy: $X^2=44.4$, $p=5.3e-9$. GCR40/GCR80: skipped (too few DEGs).

Code Block 3: Per-Tissue Fisher's Exact Test

```
fisher_mat <- matrix(c(organ_deg, organ_non, rest_deg, rest_non), nrow=2)
ft <- fisher.test(fisher_mat)
padj <- p.adjust(ft$p.value, method='bonferroni', n=5)
```

What it does: For each organ, builds a 2x2 table (organ vs rest, DEG vs non-DEG) and runs Fisher's exact test. Then applies Bonferroni correction for 5 tests (one per organ).

Statistical concept - Fisher's exact test: Fisher's exact test is an alternative to the chi-square test for 2x2 tables. It is preferred when sample sizes are small because it calculates the exact probability rather than relying on a large-sample approximation. The test returns an odds ratio (OR) and a p-value.

Intermediate Term: Bonferroni Correction

When running multiple tests, the chance of a false positive increases. The Bonferroni correction adjusts for this by multiplying each p-value by the number of tests (here, 5). If you run 5 tests at $\alpha=0.05$, the Bonferroni-adjusted threshold is $0.05/5 = 0.01$. This is conservative - it minimizes false positives but may miss true positives.

Results: In radiation_effect: Seedling OR=3.67 (enriched, padj=1.5e-30), Leaf_Shoot OR=1.94 (enriched, padj=1.7e-6), Root OR=0.56 (depleted, padj=5.8e-7). In Cs137_100cGy: Root OR=1.82 (enriched, padj=9.3e-4), Seedling OR=2.14 (enriched, padj=1.3e-3).

Code Block 4: Pairwise Fisher's Exact Tests

```
for (pr in combn(BROAD_ORGANS, 2, simplify=FALSE)) {
  mat <- matrix(c(contig[pr[1], 'yes'], contig[pr[1], 'no'],
    contig[pr[2], 'yes'], contig[pr[2], 'no']), nrow=2)
  ft <- fisher.test(mat)
  padj <- p.adjust(ft$p.value, method='bonferroni', n=10)
}
```

What it does: Runs all 10 pairwise comparisons between organs (5 choose 2 = 10 pairs) using Fisher's exact test, with Bonferroni correction for 10 tests.

Statistical concept - Multiple pairwise comparisons: The chi-square test tells us that organs differ overall, but not which specific pairs differ. Pairwise Fisher's tests identify exactly which organ pairs have significantly different DEG rates. With 5 organs, there are 10 possible pairs. The Bonferroni correction is stricter here ($n=10$) to account for the larger number of tests.

Results: radiation_effect: 7/10 pairs significant. Seedling (12.7%) > Leaf_Shoot (8.1%) > Seed_Silique (3.4%) = Flower (3.4%) > Root (3.1%). Cs137_100cGy: 5/10 significant. Cs137_10cGy: 4/10 significant.

Code Block 5: Direction-of-Response Chi-Square

```
dir_contig <- table(degs_only$predominant_broad_organ, degs_only$direction)
dir_chi <- chisq.test(dir_contig)
```

What it does: Tests whether the up/down ratio differs across organs for DEGs within a comparison.

Statistical concept - Testing conditional distributions: This is another chi-square test, but instead of testing DEG vs non-DEG, it tests up vs down. The question is: do all organs have the same up/down ratio, or do some organs respond differently in direction? A significant result means the direction of response is tissue-dependent.

Results: radiation_effect: $X^2=62.7$, $p=8.0e-13$. Root is 97% up, Seedling is 100% up, Flower is 76% up (more downregulated than others). genotype_interaction: $X^2=47.7$, $p=1.1e-9$. Root is 89% down, Seedling is 100% down - the opposite pattern from radiation_effect.

Code Block 6: Cochran-Mantel-Haenszel Test (NEW)

```
# Build 2x2xK array: condition x DEG x organ strata
rad_array <- array(0, dim=c(2, 2, 5),
  dimnames=list(c('Co60', 'Cs137'), c('no', 'yes'), BROAD_ORGANS))
cmh_rad <- mantelhaen.test(rad_array)
```

What it does: Tests whether the odds ratio between condition (Co-60 vs Cs-137) and DEG status is consistent across organ strata. Returns a common odds ratio and p-value.

Statistical concept - Stratified analysis with CMH: The Cochran-Mantel-Haenszel (CMH) test is a stratified version of the chi-square test. Instead of pooling all organs together (which would confound organ differences with radiation type differences), it analyzes each organ separately and combines the results. The 'common odds ratio' is a weighted average of the organ-specific odds ratios.

Intermediate Term: Confounding

Confounding occurs when an outside variable distorts the relationship between the variables you're studying. If Seedling has more DEGs than Root, and Co-60 happens to have more Seedling samples, a simple comparison would make Co-60 look more potent - but this could be due to the tissue difference, not the radiation type. Stratification (CMH) removes this confounding by comparing within each organ first.

Results: Co-60 vs Cs-137: common OR=0.328 ($p=1.1e-39$), meaning Cs-137 has $1/0.328 = 3.1x$ lower odds of producing a tissue-specific DEG. Cs-137 dose-response: common OR=2.846 ($p=5.6e-13$), meaning 100 cGy has 2.8x higher odds than 10 cGy.

Code Block 7: Cross-Comparison Chi-Square (NEW)

```
# Compare organ distribution between radiation_effect and genotype_interaction
genotype_mat <- rbind(organ_dist_rad, organ_dist_int)
gen_chi <- chisq.test(genotype_mat)

# Compare direction patterns
dir_mat <- rbind(dir_rad, dir_int)
dir_chi <- chisq.test(dir_mat)
```

What it does: Two chi-square tests comparing the tissue distribution and direction patterns between the radiation main effect and the sog1-1 interaction effect.

Statistical concept - Comparing distributions across conditions: These tests use a 2x5 (condition x organ) or 2x2 (condition x direction) contingency table. The first asks: does the organ distribution of DEGs differ between the two comparisons? The second asks: does the up/down ratio differ?

Results: Organ distribution: $X^2=87.1$, $p=5.5e-18$ (significantly different). Direction: $X^2=456.5$, $p=2.8e-101$ (extremely different). The direction result is one of the strongest signals in the entire analysis - the sog1-1 interaction reverses the direction of the radiation response.

Part III: Statistical Glossary

Adjusted p-value (padj)

A p-value corrected for multiple testing. DESeq2 uses the Benjamini-Hochberg procedure, which controls the false discovery rate (FDR). A padj of 0.05 means we expect 5% of called DEGs to be false positives.

Bonferroni Correction

A conservative multiple testing correction that multiplies each p-value by the number of tests. If you run 5 tests, the threshold becomes $0.05/5 = 0.01$. It minimizes false positives but may miss true positives.

Chi-Square Test of Independence

Tests whether two categorical variables are associated. Here, it tests whether organ membership and DEG status are independent. A large χ^2 statistic with a small p-value means the variables are associated (organs have different DEG rates).

Cochran-Mantel-Haenszel (CMH) Test

A stratified version of the chi-square test. Instead of pooling all data, it analyzes each stratum (organ) separately and combines results. This removes confounding by the stratifying variable. Returns a common odds ratio.

Confidence Interval (CI)

A range of values that likely contains the true parameter. A 95% CI means that if you repeated the experiment many times, 95% of the intervals would contain the true value. Narrower CIs indicate more precise estimates.

Confounding

When a third variable distorts the relationship between two studied variables. For example, if organs have different baseline DEG rates, comparing radiation types without accounting for organ would confound tissue effects with radiation effects.

Contingency Table

A table showing the frequency distribution of two or more categorical variables. Rows and columns are categories; cells are counts. The chi-square test operates on contingency tables.

DESeq2

An R package for differential expression analysis of RNA-seq count data. It models counts with a negative binomial distribution and performs internal normalization (median-of-ratios). Requires raw integer counts as input.

Effect Size

The magnitude of a difference or relationship. In differential expression, the effect size is the log2 fold change. A large effect size means a big change in expression, regardless of statistical significance.

Fisher's Exact Test

An exact test for 2x2 contingency tables. Preferred over chi-square when sample sizes are small. Returns an odds ratio and p-value. Used here for per-organ and pairwise comparisons.

Frequency Distribution

A summary of how often each value or category occurs. Bar charts visualize frequency distributions for categorical variables.

Log2 Fold Change (log2FC)

The log base 2 of the fold change. A log2FC of 1 means 2-fold upregulation; -1 means 2-fold downregulation. Log scale makes up and down changes symmetric: +1 and -1 are equally far from 0.

Negative Binomial Distribution

A probability distribution for count data that allows for overdispersion (variance greater than the mean). DESeq2 uses this to model RNA-seq read counts.

Odds Ratio (OR)

The ratio of odds between two groups. OR=2 means the event is twice as likely in group 1 vs group 2. OR=0.5 means half as likely. OR=1 means no difference. Used in Fisher's exact and CMH tests.

P-value

The probability of observing data as extreme as what you got, assuming the null hypothesis is true. A small p-value (< 0.05) suggests the null hypothesis is unlikely. It does NOT measure the probability that the null is true.

Stratification

Analyzing data within subgroups (strata) separately, then combining results. The CMH test stratifies by organ. This removes confounding by the stratifying variable and is more powerful than ignoring subgroup differences.

Tau Tissue-Specificity Index

A score from 0 to 1 measuring how specifically a gene is expressed in one tissue. 0 = expressed equally everywhere; 1 = expressed in only one tissue. Threshold of 0.6 classifies genes as tissue-specific.

Z-score

The number of standard deviations a value is from the mean. $z = (x - \text{mean}) / \text{sd}$. Used in heatmaps to normalize expression across genes with different absolute levels. Positive z = above average; negative = below average.